Assignment-4

1) What is JSP? Enlist advantages of JSP over Servlet?

→

JSP (Java Server pages) :-

- JSP is a server-side technology used to create dynamic web pages. it allows developers to embed Java code directly into HTML pages using special JSP tags.
- JSP is built on top of Servlets and is used to simplify the development of web applications by separating presentation (HTML) from business logic.
- When a JSP page is requested, the web server automatically converts it into a Servlet, compiles it and execute it to generate a response in the form of html.
- For example:

```
<html>
<body>
    <h2> Welcome <%= request.getParameter("
    name") %> </h2>
</body>
</html>
```

Here, <%= %> is used to insert java code inside HTML

Advantages of JSP over Servlet :-

① Easy to develop :- JSP is easier to write compared to Servlets because it allows embedding Java code directly within HTML.

② Better readiability:- JSP provides better readability and maintainability because the structure of the web page is clearly visible.

③ Faster Development:- JSP supports rapid development as it requires less coding effort.

④ Built in implicit objects:- JSP provides several implicit objects like request, response, session and out which can be used directly without explicit declaration.

⑤ Automatic Compilation:- JSP pages are automatically compiled into Servlets by the server then requested. So developers do not need to compile manually.

⑥ Platform independent:- Instead of being tied to single platforms, JSP technology can run on any web server and is supported by wide variety tools.

2) Draw and explain MVC architecture for developing web applications.

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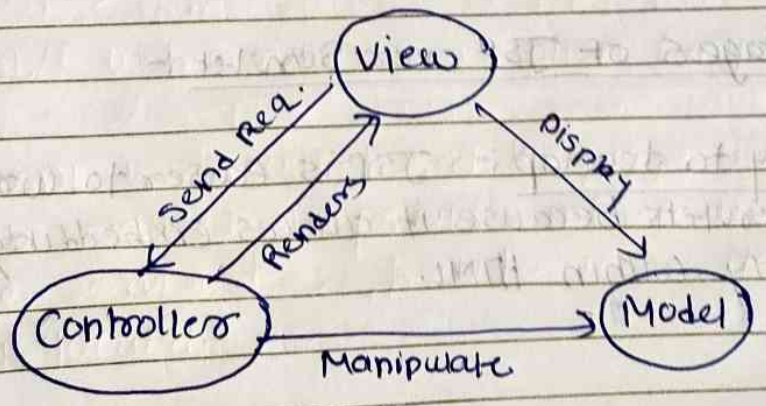

Fig. MVC Architecture

- The Model-View-Controller (MVC) is an architectural pattern that separates an application into three main logical components: the model, the view and the controller.
- Each of these components are built to handle specific deployment aspects of an application.
- MVC is one of the most frequently used industry standard web development framework to create scalable and extensible projects.
- MVC stands for Model, View and controller.

MVC separates application into three components.
Model, view and Controller.

① Model: Model represents shape of the Data and business logic. It maintains the data of the application. Model objects retrives and store model state in a dataset.

② View:- View is a user interface. View display data using model to the user and also enables them to modify the Data. View is a User Interface.

③ Controller:- Controller handles the user request. typically user interact with view, which in-turn raises appropriate URL requests, this request will be handled by a Controller. The controller renders the appropriate with the model data as a response.

- MVC is popular as it isolates the application logic from the user interface layer and supports separation of concerns.

- Here the controller receives all requests for the application and then works with the Model to prepare any data needed by the view.

3) What is WSDL and SOAP? Explain WSDL in details.

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SOAP (Simple object Access protocol):-

- SOAP is a standard protocol used for exchanging structured information between web services over a network.
- It is based on XML and provide a platform-independent way of communication between applications.
- SOAP allows programs written in different language and running on different platforms to communicate with each other.
- A SOAP message is an XML document that contains an envelope, header, and body.
- The envelope defines the start and end of the message. the header contains optional information such as authentication, and the body contains the actual data or request.

WSDL (Web Services Description Language):-

- WSDL is an XML-based language used to describe web services. It provides information about what the service does, how to access it and where it is located.

- The main structure of WSDL document look like this:

① Types: which provides data type definition used to describe the message exchanged.

Syntax:

```
<types>
        definition of types.....
</types>
```

② message:- which represents an abstract definition of the data being transmitted. A message consists of logical parts, each of which associated with a definition.

Syntax:

```
<message>
        definition of message.....
</message>
```

③ portType:- which is a set of abstract operations. Each operation refers to an input message and output message

Syntax:

```
<portType name = "Service portType">
        definition of portType....
</portType>
```

④ Binding:- which specifies concrete protocol and data format specifications for the operations and message defined by a particular portType.

Syntax:

```
<binding>
        definition of a binding.....
</binding>
```

Date
Page

⑤ Service: which is used to aggregate a set of related ports.

Syntax:

```
<service>
        definition of a service.....
</service>
```

4) Write a JSP program to demonstrate use of page directive, Scriplet Expression and Comment

→

```
<%@ page import = "java.util.Date" contentType =
" text/ html" %>

<html>
<head>
    <title> JSP Demo </title>
</head>
<body>
  <!-- HTML Comment -->

  <%-- JSP Comment : this will not be visible in
browser -- %>

  <h2> Demonstration of JSP elements </h2>
<%
      // Scriplet : Java code execution
        String name = "Anilet";
        int a=10 , b = 20;
        int sum = a+b;
%>
```

```
<p> Hello, <%= name %> </p> <!-- Expression -->

<p> Sum of two number is : <%= sum %> </p>
  <!-- Expression -->

<p> Current Date and Time : <%= new Date() %>
</p> <!-- Expression -->

</body>
</html>
```

5) List & describe important inceptors provided by Strut framework.

→ The Strut framework provides a comprehensive set of pre-defined interceptors and common interceptors stacks which are given below:

① Alias:- It converts similar parameters that have different names between requests.

② Checkbox:- It is used to handle the check boxes in the form. By this we can detect the ~~unch~~ unchecked checkboxes.

③ Cookie:- It adds a cookie to the current Action.

④ ConversionError:- It adds conversion error to the action's field errors.

⑤ CreateSession:- It creates an HttpSession object if it doesn't exists.

(6) clearSession : It unbinds the HttpSession object.

(7) exception: It maps exception to a result

(8) fileUpload :- It provides support to file upload in struts2.

(9) i18n :- It provides support to internationalization and localization.

(10) logger: it outputs the action name.

(11) Store : It stores and retrive action messages, action errors ~~to~~ or field errors for action that implements ValidationAwareInterface.

(12) modelDriven :- It makes other model object as the default object of valuestack.

(13) params :- it populates the action properties with the request parameters.

(14) Scope : - It is used to store the action state in the session or application scope.

(15) validation :- it provides support to input validation.